

Data Flow Diagrams & User Stories

Data Flow Diagram (DFD) description

Data originates from Kaggle as a raw CSV file. It is then imported into Tableau where calculated fields and aggregations are applied. The processed data is visualized across 8 individual worksheets, which are finally consolidated into an interactive Dashboard and a 3-scene Storyboard. The Tableau workbook is published to Tableau Public, and the resulting public URL is embedded into a custom Flask web application UI.

```
graph TD;
    Kaggle_Dataset-->|CSV Download|Local_Storage;
    Local_Storage-->|Import|Tableau_Data_Source;
    Tableau_Data_Source-->|Data Prep & Joins|Tableau_Workbooks;
    Tableau_Workbooks-->|Assemble 8 Charts|Tableau_Dashboard;
    Tableau_Workbooks-->|Curate Insights|Tableau_Storyboard;
    Tableau_Dashboard-->|Publish|Tableau_Public;
    Tableau_Storyboard-->|Publish|Tableau_Public;
    Tableau_Public-->|Embed URL|Flask_Web_App;
    Flask_Web_App-->Store_Manager;
```

User Stories

- **As a** Store Manager, **I want to** see the Average Sales Volume by Product Category in a web browser, **so that I can** identify our top-performing product types without needing Tableau installed.

- **As a** Merchandiser, **I want to** compare Competitor Price vs Price by category using interactive filters, **so that I can** ensure our pricing strategy remains competitive.
- **As a** Retail Executive, **I want to** view a 3-scene Tableau Storyboard embedded in our internal portal, **so that I can** quickly grasp key insights regarding product placement.
- **As a** Tableau Developer, **I want to** publish the finalized dashboard to Tableau Public, **so that I can** securely embed it into a Flask web application using HTML iframe/script tags.